# From Shadow Lengths to Density Matrices: A No-Physics Route to a Finite-Dimensional Gleason-Type Theorem

Repository Example for a Theoretical Workflow

**Abstract**

Gleason's original theorem is deep, important, and notoriously hard to explain to beginners. This note does not claim a new elementary proof of the original 1957 theorem. Instead, it isolates a simpler target that still captures the density-matrix conclusion in finite dimension. We prove a finite-dimensional real Gleason-type theorem for additive rules on the effect space of symmetric matrices. The proof is built in three steps. First, we prove an interval lemma: every bounded additive rule on $[0, 1]$ is linear. Second, we turn that lemma into a high-school-friendly sphere corollary for rules that depend only on squared height. Third, we lift the same additivity idea to real symmetric matrices and show that every normalized additive rule on effects has the form $E \mapsto \mathrm{tr}(\rho E)$ for a unique positive semidefinite trace-one matrix $\rho$. The note avoids quantum-mechanics vocabulary and keeps the proof in bounded additivity, orthogonality, and finite-dimensional linear algebra. It also states the gap to the original theorem explicitly: the argument is finite-dimensional, real, and assumes additivity on the full effect space rather than only on orthogonal projections. To satisfy the repository's reproducibility standard, we also ship generated numerical sanity checks and high-resolution figure assets. The computations do not replace the proof, but they do confirm that the reconstruction formulas behave at machine precision on random sampled examples. The result should therefore be read as an entry point into Gleason-style rigidity rather than as a substitute for the classical theorem.

## 1 Introduction

Gleason's theorem is one of the structural pillars behind the density-matrix form of quantum probability. It is also a theorem that beginners often find forbidding. Even so-called elementary proofs still demand a fair amount of geometric or analytic maturity (Cooke et al., 1985). The goal of this note is not to compete with the original theorem or with the best existing proofs. The goal is more modest and, we think, more teachable.

We give a short route to a *finite-dimensional real Gleason-type theorem*. The theorem concerns additive rules on the effect space

$$\mathcal{E}_d = \{E \in \mathbb{R}^{d\times d}_{\mathrm{sym}} : 0 \le E \le I\}.$$

Here $0 \le E \le I$ means that $E$ is positive semidefinite and all its eigenvalues are at most 1. A rule $f : \mathcal{E}_d \to [0, 1]$ is assumed to satisfy $f(I) = 1$ and

$$f(E + F) = f(E) + f(F)$$

whenever $E$, $F$, and $E + F$ all lie in the same effect space. The conclusion is that $f(E) = \mathrm{tr}(\rho E)$ for a unique positive semidefinite trace-one matrix $\rho$.

This statement is not Gleason's original theorem. The assumption is stronger: additivity is required on the whole effect space, not merely on orthogonal projections. In that sense the note sits closer to the finite-dimensional effect-space viewpoint emphasized by Busch (2003) and discussed by Wright and Weigert

| Feature | Original Gleason theorem | This note |
|---|---|---|
| Domain | Orthogonal projections on Hilbert space | Full effect space of real symmetric matrices |
| Scalars | Full complex Hilbert-space setting | Finite-dimensional real symmetric matrices |
| Additivity input | Only on orthogonal decompositions of projections | On every decomposition $E + F$ that stays in the effect space |
| Conclusion | Density-operator rule on projections | Trace-form rule on all effects |
| Teaching cost | Substantial geometric and analytic overhead | Bounded additivity and matrix algebra |

Table 1: This note is a Gleason-type doorway rather than a replacement for the original theorem.

(2019). What is new here is not the theorem itself, but the route: a proof staircase that begins with a one-variable lemma and keeps the ideas concrete all the way through.

For readers meeting this topic for the first time, it helps to keep the comparison with the original theorem explicit. There are three simplifications at once: this note stays in finite dimension, works over real symmetric matrices rather than the full complex setting, and assumes additivity on all effects instead of only on projection-valued decompositions. The pedagogical gain is that every step can be stated in everyday linear-algebra language. The cost is that the theorem proved here is deliberately narrower.

**Contributions.** This note makes three contributions.

1. It gives a bounded-additivity warmup theorem on $[0, 1]$ that can be explained without physics.
2. It turns that warmup into a sphere corollary that is accessible with only squared lengths and orthonormal triples.
3. It uses the same bounded-additivity idea to prove the finite-dimensional real effect-space theorem in explicit matrix language.

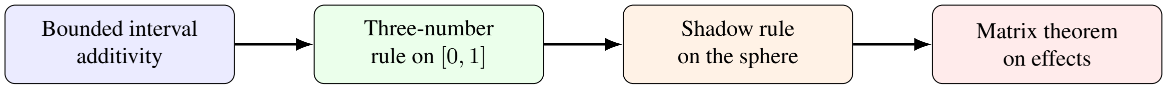
**Reader roadmap.** Readers who mainly want the intuitive core can stop after Section 2 and still see the engine of the argument. Section 3 shows how the same bounded-additivity idea becomes a matrix trace rule. Section 4 is a computational audit of the formulas rather than part of the proof. The appendix returns to plain language and explains why the original projection-only theorem remains harder.

Relative to the elementary proof tradition represented by Cooke et al. (1985), the gain here is not theorem strength but entry cost. We lower the entry threshold by changing the target theorem and making each proof step visibly local. Relative to the effect-space presentations emphasized by Busch (2003) and Wright and Weigert (2019), the contribution is again not a new mathematical statement; it is the proof staircase itself. The note is trying to make the trace-form conclusion feel teachable to a newcomer before the full classical theorem is reintroduced.

The computational artifacts in this package are intentionally secondary. They do not prove the theorem. They simply generate real reproducible checks and paper-side assets so the package meets the repo's standard for a complete final paper.

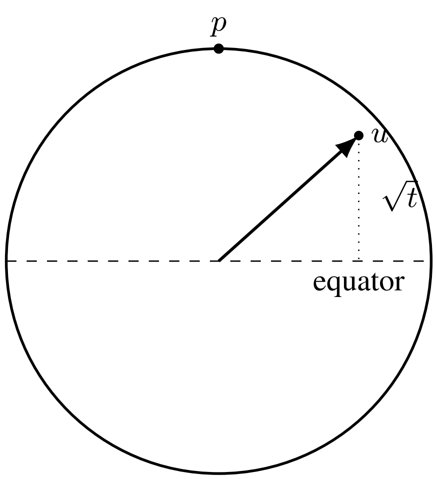
## 2 Warmup: The Interval and the Sphere

The main paper theorem will come from the same idea twice: bounded additivity forces linearity. We start in one variable because that version is simple enough to explain to a strong high-school student.




**Idea.** The paper climbs one rung at a time. First show that a bounded rule with $h(x + y) = h(x) + h(y)$ must be linear. Then turn that into an elementary sphere result. Finally lift the same additivity idea to real symmetric matrices and recover the trace formula.

Figure 1: The proof route used in the paper. Each step keeps the ideas concrete and postpones the matrix theorem until the one-variable and sphere warmups are already in place.




**Warmup picture.** If a rule depends only on the squared height $t = (u \cdot p)^2$, then an orthonormal triple gives three numbers $a, b, c$ with $a+b+c = 1$. The proof becomes a one-variable problem on $[0, 1]$.

Figure 2: The high-school warmup picture. A rule that depends only on squared height turns an orthonormal triple on the sphere into a three-number identity on $[0, 1]$.

**Lemma 1** (Bounded interval additivity)**.** *Let $h : [0, 1] \to \mathbb{R}$ be bounded. Suppose that*

$$h(x + y) = h(x) + h(y)$$

*whenever $x, y \geq 0$ and $x + y \leq 1$. Then $h(t) = ct$ for some constant $c$ and all $t \in [0, 1]$.*

*Proof.* First, $h(0) = 0$ because $h(0) = h(0 + 0) = h(0) + h(0)$.

For every positive integer $n$,

$$h(1) = h\Big(\underbrace{\tfrac{1}{n} + \cdots + \tfrac{1}{n}}_{n \text{ times}}\Big) = n\, h(1/n),$$

so $h(1/n) = h(1)/n$. Therefore, for every rational number $m/n \in [0, 1]$,

$$h(m/n) = m\, h(1/n) = (m/n)\, h(1).$$

Now use boundedness. Let $|h(t)| \leq M$ on $[0, 1]$. If $0 < u \leq 1/n$, then

$$n|h(u)| = |h(nu)| \leq M,$$

so $|h(u)| \le M/n$. Thus $h(u) \to 0$ as $u \to 0^+$.

Take any real $x \in [0,1]$ and choose rationals $q_k \to x$. Then

$$h(x) - h(q_k) = h(x - q_k)$$

when $q_k \le x$, and the right-hand side tends to 0. Since $h(q_k) = q_k h(1)$ for rational $q_k$, it follows that

$$h(x) = x\, h(1).$$

So $h(t) = ct$ with $c = h(1)$. □

**Theorem 1** (Three-number rule)**.** *Let $g : [0,1] \to \mathbb{R}$ be bounded. Suppose that*

$$g(a) + g(b) + g(c) = 1$$

*whenever $a, b, c \in [0,1]$ and $a + b + c = 1$. Then $g$ is affine:*

$$g(t) = \beta + \alpha t$$

*for constants $\alpha, \beta$ satisfying $\alpha + 3\beta = 1$.*

*Proof.* Let $\beta = g(0)$ and define $h(t) = g(t) - \beta$. Compare the two valid triples

$$(x, y, 1 - x - y) \quad \text{and} \quad (x + y, 0, 1 - x - y).$$

Subtracting the corresponding identities gives

$$h(x + y) = h(x) + h(y)$$

whenever $x + y \le 1$. Since $h$ is bounded, the lemma implies $h(t) = \alpha t$ for some $\alpha$. Hence $g(t) = \beta + \alpha t$.

Finally, plugging $(1, 0, 0)$ into the hypothesis gives

$$g(1) + 2g(0) = 1,$$

which is exactly $\alpha + 3\beta = 1$. □

**Corollary 1** (Squared-height sphere rule)**.** *Fix a unit vector $p \in \mathbb{R}^3$. Suppose $F$ assigns a real number to each unit vector $u \in S^2$ and depends only on the squared height relative to $p$:*

$$F(u) = g((u \cdot p)^2).$$

*If*

$$F(u) + F(v) + F(w) = 1$$

*for every orthonormal triple $(u, v, w)$ in $\mathbb{R}^3$, then*

$$F(u) = \beta + \alpha(u \cdot p)^2$$

*with $\alpha + 3\beta = 1$.*

*Proof.* For every orthonormal triple $(u, v, w)$,

$$(u \cdot p)^2 + (v \cdot p)^2 + (w \cdot p)^2 = \|p\|^2 = 1,$$

because $(u, v, w)$ is an orthonormal basis of $\mathbb{R}^3$. So the three numbers

$$a = (u \cdot p)^2, \quad b = (v \cdot p)^2, \quad c = (w \cdot p)^2$$

always satisfy $a + b + c = 1$. The theorem applies directly to $g$. □

The sphere corollary is not the full Gleason problem. It is a warmup. But it already captures the central pattern: a rule constrained on orthogonal triples becomes rigid because the underlying additivity is bounded.

## 3 A Finite-Dimensional Real Gleason-Type Theorem

**Theorem 2** (Real effect-space theorem)**.** *Fix $d \geq 2$. Let $\mathcal{E}_d$ be the set of all real symmetric $d \times d$ matrices $E$ satisfying $0 \leq E \leq I$. Suppose*

$$f : \mathcal{E}_d \to [0, 1]$$

*satisfies*

$$f(I) = 1$$

*and*

$$f(E + F) = f(E) + f(F)$$

*whenever $E, F, E + F \in \mathcal{E}_d$. Then there exists a unique real symmetric positive semidefinite matrix $\rho$ with $\mathrm{tr}(\rho) = 1$ such that*

$$f(E) = \mathrm{tr}(\rho E) \qquad \textit{for every } E \in \mathcal{E}_d.$$

*Proof.* We break the proof into four simple steps.

**Step 1: homogeneity on each ray.** Fix an effect $E$. The map

$$h_E(t) = f(tE), \qquad t \in [0, 1],$$

is bounded, and it satisfies

$$h_E(x + y) = h_E(x) + h_E(y)$$

whenever $x + y \leq 1$, because then $(x + y)E$, $xE$, and $yE$ all remain effects. By the interval lemma,

$$f(tE) = t\, f(E) \qquad (0 \leq t \leq 1).$$

**Step 2: extend from effects to all positive semidefinite matrices.** Let $A$ be any positive semidefinite real symmetric matrix. Choose a positive number $s$ large enough that $A/s \in \mathcal{E}_d$; for instance, any $s$ at least as large as the largest eigenvalue of $A$ will do. Define

$$F(A) = s\, f(A/s).$$

This does not depend on the chosen $s$. Indeed, if $A/s$ and $A/t$ are both effects and $s \leq t$, then

$$A/t = (s/t)(A/s),$$

so Step 1 gives

$$t\, f(A/t) = t\,(s/t)\, f(A/s) = s\, f(A/s).$$

Thus $F$ is well defined.

Now let $A, B$ be positive semidefinite. Choose $u$ large enough that $(A + B)/u$ is an effect. Then

$$F(A + B) = u\, f((A + B)/u) = u\, f(A/u) + u\, f(B/u) = F(A) + F(B).$$

So $F$ is additive on the positive semidefinite cone.

**Step 3: extend to all symmetric matrices.** Every real symmetric matrix $H$ can be written as

$$H = A - B$$

with $A, B$ positive semidefinite. Define

$$L(H) = F(A) - F(B).$$

This is well defined. If $H = C - D$ also, then $A + D = B + C$, and additivity of $F$ gives

$$F(A) + F(D) = F(B) + F(C),$$

so $F(A) - F(B) = F(C) - F(D)$.

Linearity of $L$ is now immediate:

$$L((A - B) + (C - D)) = F(A + C) - F(B + D) = L(A - B) + L(C - D),$$

and $L(\lambda H) = \lambda L(H)$ follows from the positive-scalar homogeneity of $F$.

**Step 4: write $L$ as a trace rule.** Let $e_1, \dots, e_d$ be the standard basis vectors. Define

$$P_i = e_i e_i^\top$$

and, for $i < j$,

$$S_{ij} = \tfrac{1}{2}(e_i + e_j)(e_i + e_j)^\top.$$

Each $P_i$ and $S_{ij}$ is an effect; in fact, $S_{ij}$ is the rank-one projection onto the unit vector $(e_i + e_j)/\sqrt{2}$. Set

$$\rho_{ii} = f(P_i), \qquad \rho_{ij} = f(S_{ij}) - \tfrac{1}{2}f(P_i) - \tfrac{1}{2}f(P_j) \quad (i < j),$$

and define $\rho_{ji} = \rho_{ij}$. This gives a real symmetric matrix $\rho$.

Now let

$$X_{ij} = \tfrac{1}{2}(E_{ij} + E_{ji}) = S_{ij} - \tfrac{1}{2}P_i - \tfrac{1}{2}P_j.$$

So $X_{ij}$ is exactly the symmetric off-diagonal direction recovered from values of $f$ on honest effects. By construction,

$$L(P_i) = \rho_{ii}, \qquad L(X_{ij}) = \rho_{ij}.$$

Every real symmetric matrix $H = (h_{ij})$ has the basis expansion

$$H = \sum_{i=1}^{d} h_{ii} P_i + 2 \sum_{1 \le i < j \le d} h_{ij} X_{ij}.$$

Therefore,

$$L(H) = \sum_{i=1}^{d} h_{ii}\rho_{ii} + 2 \sum_{1 \le i < j \le d} h_{ij}\rho_{ij} = \operatorname{tr}(\rho H).$$

We still need positivity and normalization. If $v$ is any vector, then $vv^\top$ is positive semidefinite, so

$$v^\top \rho v = \operatorname{tr}(\rho v v^\top) = L(vv^\top) = F(vv^\top) \ge 0.$$

Hence $\rho$ is positive semidefinite.

Table 2: Generated numeric sanity checks for the theorem package. The theorem is proved analytically; the table simply confirms that the basis-effect reconstruction and additivity formulas behave at machine precision on random sampled examples.

| $d$ | Max recon. error | Max rep. residual | Max additivity residual |
|---|---|---|---|
| 2 | 1.665e-16 | 2.220e-16 | 2.220e-16 |
| 3 | 1.110e-16 | 1.110e-16 | 1.110e-16 |
| 4 | 1.110e-16 | 1.110e-16 | 5.551e-17 |

Also,

$$I = P_1 + \cdots + P_d,$$

and the partial sums remain effects, so

$$\mathrm{tr}(\rho) = \sum_{i=1}^{d} \rho_{ii} = \sum_{i=1}^{d} f(P_i) = f(I) = 1.$$

Finally, if $E \in \mathcal{E}_d$, then $E$ itself is positive semidefinite, so

$$f(E) = F(E) = L(E) = \mathrm{tr}(\rho E).$$

Uniqueness is immediate because the values on the basis effects determine every entry of $\rho$. $\square$

The argument above is intentionally concrete. No Hilbert-space lattice, no measure theory, and no quantum postulates are needed. The only real inputs are bounded additivity on $[0, 1]$, positive semidefinite decompositions, and a basis expansion of symmetric matrices.

# 4 Generated Numeric Checks

The theorem is analytic, not experimental, so no amount of numerics can prove it. Even so, this repository requires a complete paper package to ship with executable artifacts. For that reason, the note includes a small computation layer that does two reproducible jobs.

First, it samples random positive semidefinite trace-one matrices $\rho$ in dimensions 2, 3, and 4. From the theorem's basis formulas, it reconstructs the matrix entries of $\rho$ using only values of the state rule on the effects $P_i$ and $S_{ij}$. Second, it checks the recovered matrix on random effects and on random additive decompositions of effects.

Table 2 shows the outcome. Across 250 trials per dimension, the reconstruction errors, representation residuals, and additivity residuals remain at machine scale. That is exactly what should happen if the paper formulas are implemented correctly.

The package also generates sphere checks for the warmup corollary by sampling random orthonormal triples and verifying that the affine squared-height rule sums to 1 up to floating-point tolerance. Again, these checks are illustrative, not evidentiary. Their role is to make the theorem package computationally inspectable rather than magical.

That role is pedagogically useful. A beginner can now look at the basis formulas, run the code, and watch the claimed trace representation reappear numerically from finitely many evaluations of the rule. The computation still proves nothing, but it does remove one source of intimidation: the reader no longer has to take the reconstruction formulas on faith.

## 5 Discussion and Limits

This note is deliberately careful about what it does not prove.

The cleanest way to state the gap is as a three-part checklist. First, the theorem here assumes additivity on the full effect space rather than only on orthogonal projections. Second, it stays in the real symmetric setting rather than the full complex Hilbert-space statement in Gleason (1957). Third, it is a finite-dimensional theorem. In that sense the note sits much closer to the effect-space perspective emphasized by Busch (2003) and discussed cleanly by Wright and Weigert (2019) than to the original projection-only theorem.

It also does not claim that the effect-space theorem proved here is new as mathematics. The mathematical contribution is expository and structural: the point is to make the route to the trace form feel inevitable to a newcomer rather than miraculous on first contact.

What the note does contribute is a more teachable route. The proof starts from a one-variable bounded-additivity lemma, turns that into a sphere warmup that can be explained with squared heights, and only then moves to real matrices. That progression matters if the goal is pedagogy, onboarding, or conceptual demystification rather than maximal theorem strength.

There is also a limit to the computational layer. The generated checks confirm that the reconstruction formulas and state-rule identities behave correctly in random sampled examples. They do not strengthen the theorem mathematically. In a theoretical paper, the proof remains primary.

The clearest next step would be a follow-on note that closes one part of the gap at a time: first by recovering more of the projection-only setting, then by clarifying which parts of the argument survive unchanged in the complex case, while preserving as much of the present accessibility as possible.

## 6 Conclusion

The right way to make a hard theorem accessible is not always to flatten the full theorem at once. Sometimes the better route is to separate the core idea from the heaviest machinery and rebuild the argument in smaller pieces.

That is the philosophy of this note. The high-school warmup explains why bounded additivity forces linearity. The sphere corollary shows how orthogonality turns geometry into a one-variable identity. The matrix theorem then packages the same idea into the trace formula

$$f(E) = \mathrm{tr}(\rho E).$$

The result is not a replacement for Gleason's original theorem. It is a cleaner doorway into the density-matrix conclusion and the type of rigidity that makes Gleason-style results powerful in the first place.

That distinction is not a weakness to be hidden. It is the paper's organizing discipline. A reader who finishes this note should know both what has been proved and exactly what remains to be added before one returns to the original projection-only theorem. For an expository contribution, that kind of honesty is part of the result.

## A Plain-Language Note

The first half of the paper can be summarized without matrices.

Suppose you have a rule that attaches a number to every point on a line segment from $0$ to $1$. If the number for a combined piece is always the sum of the numbers for the smaller pieces, and if the rule never blows up to infinity, then the rule has to be a straight line. That is the whole engine of the proof.

On the sphere, squared heights of an orthonormal triple always add to $1$. So a rule that only depends on squared height is forced into the same straight-line pattern. The matrix theorem is a larger version of the same story: once additivity is available on enough positive pieces, the whole rule becomes linear and must come from one fixed matrix.

### A.1 Why the original theorem is harder

This note simplifies the classical Gleason problem in two especially important ways.

First, it assumes additivity on the full effect space rather than only on projections. That matters because the proof repeatedly scales an effect along a ray and writes one effect as a sum of smaller effects. For projections alone, that maneuver disappears: if $P$ is a projection, then $tP$ is usually *not* a projection unless $t = 0$ or $1$. So the easy interval-lemma route is no longer available for free.

Second, the note stays in the real symmetric setting. In the full complex case, off-diagonal directions carry both real and imaginary parts, and the bookkeeping needed to reconstruct the representing operator is correspondingly richer. The present proof avoids that extra layer on purpose so the reader can see the core rigidity idea before facing the full theorem.

## References

Paul Busch. Quantum states and generalized observables: A simple proof of Gleason's theorem. *Physical Review Letters*, 91(12):120403, 2003.

Roger Cooke, Michael Keane, and W. Moran. An elementary proof of Gleason's theorem. *Mathematical Proceedings of the Cambridge Philosophical Society*, 98(1):117–128, 1985.

Andrew M. Gleason. Measures on the closed subspaces of a hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.

John D. M. Wright and Stefan Weigert. Gleason-Type Theorems from Cauchy's Functional Equation. *Foundations of Physics*, 49(7):696–708, 2019.